System setup: This is essentially the first step. You often need assistance to figure out how many nodes are needed to query across a certain amount of data. That is when auto provisioning comes into the picture. You have to load your data, and the auto provisioning advisor will tell how many nodes you require based on the number of tables and primary keys.

Data load: The auto data load advisor consists of auto parallel loading, auto data placement, and auto encoding. Auto parallel loading is designed intelligently to tell you the number of threads you require to load a certain piece of data in parallel. Auto data placement predicts the column on which the table should be partitioned in the memory in order to achieve the best performance for queries. It also predicts the expected gain in query performance with a new column recommendation.

Query execution: This consists of auto scheduling, auto change propagation, auto query time estimation, and auto query plan improvement. Here again the query plan advisor simply executes a query and improves the execution plan for future queries. It works in an efficient way to optimise query plans via the machine learning algorithms. This allows real-time optimisation of queries because the query optimiser learns and adapts during the database usage time. Auto query time estimation is based on an analytical model of Autopilot's query processing algorithms. Using a data driven approach improves performance over time as more queries are run. In auto scheduling, short queries are prioritised over long running queries. The system reduces the wait time for shorter queries without changing total execution time. Auto change propagation intelligently determines the optimal time for changes in the MySQL database to be propagated.

Failure handling: This consists of auto error recovery. Whenever there is a HeatWave error, the recovery is automatically taken care of as it is being continuously monitored by the Autopilot advisor. A system failure is handled by restarting the failed node and the HeatWave error is fixed by restarting the failed HeatWave process.

To wrap things up, it is essential to note that MySQL Autopilot supports machine learning based automation. It provides functionalities such as data loading, query execution and failure handling while further improving the performance of HeatWave itself. It is faster and more economical than its competitors. However, remember to make a clear assessment of the requirements before choosing a service as the system's productivity will rely heavily upon that. END

MySQL Enterprise Edition: A fully managed MySQL database service

MySQL Enterprise Edition comes with a whole lot of functionalities such as backup, replication, monitoring and many other tools associated with it. Any on-premise customer using this version of MySQL does not have to worry about the latest updates for their databases as all new updates to the Enterprise Edition are added automatically. This not only saves time but also lets customers focus more on their applications than on maintaining the database and its version.

	Automation	MySQL On-Premises	MySQL Database Service
Database	Scaling	✗	✓
	Backup	✗	✓
	Security Patch & Upgrade	✗	✓
	Provision & Configure	✗	✓
OS	OS Security Patch & Upgrade	✗	✓
	OS Installation	✗	✓
Server	Hardware Purchase & Maintenance	✗	✓
Storage	Storage Purchase & Maintenance	✗	✓
Data Center	Rack & Space	✗	✓
	Power, HVAC, Networking	✗	✓

Figure 4: A fully managed MySQL database service

By: Jothir Ganesan

The author is the site reliability engineering development manager at Oracle. This article is based on the tech talk he gave at OSI 2021.

Transcribed and curated by Laveesh Kocher